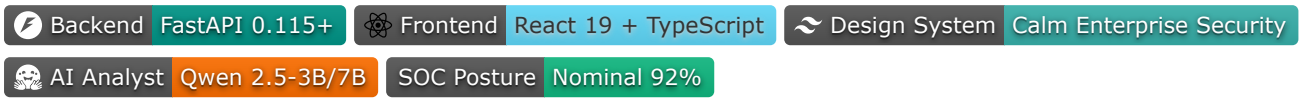
CYBERGUARD — AI-Powered Cyber Threat, Phishing & Digital Impersonation Defense Platform



> **Built for the Odisha Knowledge Corporation Limited (OKCL) Hackathon — Problem Statement 9:** *"CYBERGUARD - AI Powered Cyber Threat, Phishing & Digital Impersonation Detection and Response System"*

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1. Executive Overview & Problem Statement

Modern state and educational infrastructures face attacks that target two distinct planes: - **The Technology Layer:** Port sweeps, SYN floods, credential stuffing, SSH password spraying, API abuse. - **The Human Layer:** Spoofed SMS notifications (Smishing), spear-phishing emails (BEC), QR payment traps (Quishing), cloned executive voices (Voice Deepfakes), and synthetic keynote video manipulation.

Traditional firewalls only monitor network packets, remaining blind to human-targeted multimodal deception. **CyberGuard** bridges this gap by unifying network telemetry with computer vision, acoustic spectral processing, graph analytics, and explainable generative AI into a single autonomous Security Operations Center (SOC) platform.

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2. Core Architectural Principle

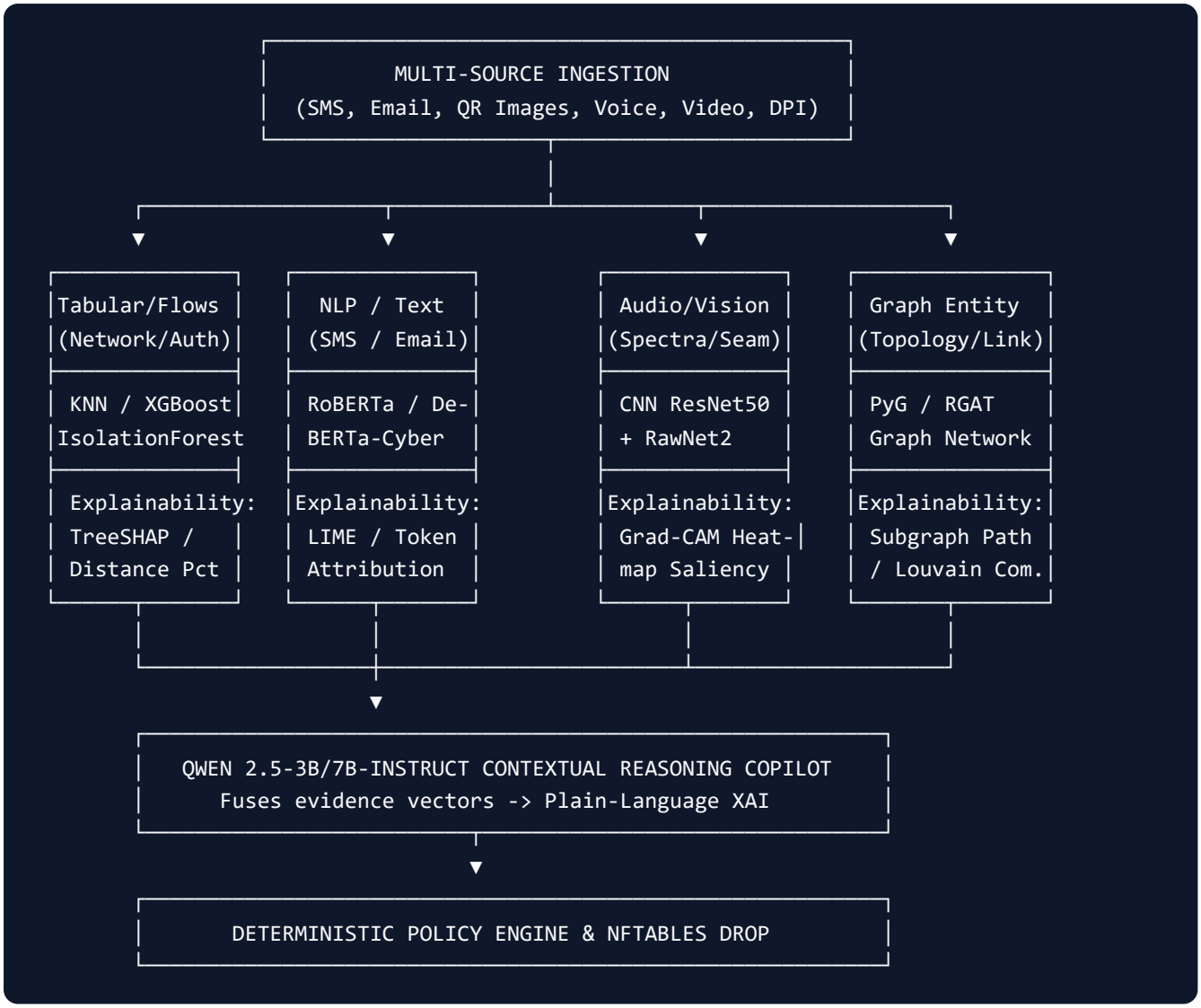
```
AI ADVISES  ──►  DETERMINISTIC POLICY ENFORCES
```

> [!IMPORTANT] **Why we do NOT allow Large Language Models to directly execute firewall commands:** LLMs are probabilistic, prone to hallucinations, and vulnerable to prompt injection. In CyberGuard: 1. **AI Models (KNN, CNN, Transformers, GNN, Qwen 2.5)** inspect, classify, score, correlate, and explain threats. 2. **Deterministic Policy Engine** evaluates mathematical risk thresholds, verifies constraints, synthesizes temporary `nftables` rules with TTL expiration, and verifies kernel-level Netlink receipts.

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3. Multi-Model AI Subsystem Architecture

A single neural network cannot solve text, audio, images, and network packets simultaneously. CyberGuard uses a **Hierarchical Multi-Model AI Ensemble**:


MULTI-SOURCE INGESTION
(SMS, Email, QR Images, Voice, Video, DPI)
Tabular/Flows (Network/Auth)
KNN / XGBoost IsolationForest
Explainability: TreeSHAP / Distance Pct
NLP / Text (SMS / Email)
RoBERTa / De-BERTa-Cyber
Explainability: LIME / Token Attribution
Audio/Vision (Spectra/Seam)
CNN ResNet50 + RawNet2
Explainability: Grad-CAM Heat-map Saliency
Graph Entity (Topology/Link)
PyG / RGAT Graph Network
Explainability: Subgraph Path / Louvain Com.
QWEN 2.5-3B/7B-INSTRUCT CONTEXTUAL REASONING COPILOT
Fuses evidence vectors -> Plain-Language XAI
DETERMINISTIC POLICY ENGINE & NFTABLES DROP

Specialized Model Breakdown:

Model Architecture	Targeted Threat Vector	Input Features	Explainability Engine (XAI)
KNN & Isolation Forest	Network bursts & Geo-Velocity Impossible Travel	Flow inter-arrival rates, SYN/ACK ratios, $(\Delta \text{distance}, \Delta \text{time})$	Euclidean σ-distance from authentic centroid clusters
Spectral CNN (RawNet2)	Audio / Voice Deepfakes	128-band Mel-Spectrograms, Constant-Q Transforms	Vocoder roll-off ($>8.1\text{ kHz}$) & F0 pitch contour flatlines
ResNet50 / ViT (CNN)	Image Quishing & Video Facial Seam Deepfakes	224x224 RGB crops & DCT frequency transforms	Grad-CAM saliency heatmaps highlighting tampered facial boundaries
Fine-Tuned RoBERTa	SMS Smishing & Phishing Emails	Token sequences, Telecom DLT header metadata	Attention attribution highlighting psychological urgency tokens
Heterogeneous GNN (PyG)	Multi-Hop Attack Infrastructure & Lateral Movement	Node embeddings (Actor, Domain, IP, SMS, Victim)	GNNExplainer identifying minimum connected threat subgraphs
Qwen 2.5-3B/7B-Instruct	Incident Synthesis & Human-in-the-Loop Co-pilot	Consolidated multi-modal telemetry + Suricata logs	Evidence-grounded natural language justifications + MITRE mapping

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4. AI Processing & Feature Extraction Pipeline

A. SMS & Smishing Pipeline (`backend/main.py`)

1. **Telecom DLT Header Verification:** Checks sender ID against the official telecom regulatory registry (flags unauthorized headers such as `OKCL-GOV`, `OKCL-STUDENT`).
2. **Urgency & Coercion NLP Classifier:** Scores psychological pressure phrases ("*within 2 hours*", "*immediate de-registration*", "*bank KYC*").
3. **URL Evasion Extraction:** Extracts embedded shortlinks and measures Shannon entropy.

B. Image & QR Quishing Pipeline (`components/multimodal/QRCodeInspector.tsx`)

1. **Matrix Decoding:** Decodes concealed QR barcode payloads using OpenCV / pyzbar.
2. **Computer Vision OCR:** Tesseract OCR extracts surrounding notice text from promotional posters.

3. **Destination Reputation:** Analyzes domain age (e.g. registered 3 days ago), free Let's Encrypt SSL certificates, and typosquatting.

C. Audio & Voice Deepfake Pipeline (`components/multimodal/AudioWaveformVisualizer.tsx`)

1. **Acoustic Spectrogram Transform:** Audio converted to 128-band Log-Mel Spectrogram.
2. **Neural Vocoder Detection:** Detects sharp high-frequency roll-off above $8,150\text{ Hz}$ typical of HiFi-GAN and DiffWave synthesizers.
3. **Prosody Variance (F_0):** Measures pitch inflection; cloned voices exhibit robotic flatlines ($\sigma < 5\text{ Hz}$).
4. **Biometric Distance:** Cosine distance comparison against enrolled executive voice embeddings.

D. Video Deepfake Pipeline (`components/multimodal/VideoFrameAnalyzer.tsx`)

1. **Facial Seam Extraction:** Evaluates 120 timeline frames for Gaussian boundary artifacts around the jawline.
2. **Blinking Rate Frequency:** Flags abnormal eye-aspect ratio rates (e.g., 1.2 blinks/min vs. natural human 14 blinks/min).
3. **Phoneme-to-Viseme Synchronization:** Flags temporal phase lag between spoken audio and lip movements.

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5. End-to-End API Reference & Frontend Bindings

All endpoints are hosted on `http://127.0.0.1:8000/api/v1` and consumed by `frontend/src/services/api.ts`:

Endpoint	Method	Purpose	Frontend Component / Page
`/auth/login`	`POST`	Authenticate with Argon2id & return JWT pair	`LoginPage.tsx`
`/auth/logout`	`POST`	Revoke active refresh token	`Topbar.tsx`
`/me`	`GET`	Current user profile, role, tenant	`AuthContext.tsx`
`/dashboard/summary`	`GET`	Posture score, metrics, vector breakdown	`OverviewPage.tsx`
`/threats/live`	`GET`	Ingress threats stream with Suricata alerts	`LiveThreatsPage.tsx`
`/threats/block`	`POST`	1-Click manual IP block	`LiveThreatsPage.tsx`
`/firewall/rules`	`GET`	Active nftables rules inventory with drop counters	`AdaptiveFirewallPage.tsx`
`/firewall/toggle`	`POST`	Enable / Disable specific firewall rule	`AdaptiveFirewallPage.tsx`
`/multimodal/records`	`GET`	Ingested SMS, Email, QR, Voice, Video records	`MultimodalThreatLabPage.tsx`
`/multimodal/analyze`	`POST`	Run AI forensic scan on uploaded content	`MultimodalThreatLabPage.tsx`
`/graph/topology`	`GET`	Heterogeneous attack graph nodes and edges	`GraphIntelligencePage.tsx`
`/ai/chat`	`POST`	Evidence-grounded Qwen 2.5 security co-pilot	`AIAnalystPage.tsx`
`/simulator/scenarios`	`GET`	Runnable attack scenarios	`SimulatorPage.tsx`
`/simulator/run`	`POST`	Execute attack simulation in safe sandbox	`SimulatorPage.tsx`
`/responses`	`GET`	Netlink verified response receipts	`ResponseCenterPage.tsx`
`/responses/execute`	`POST`	Execute manual mitigation receipt	`ResponseCenterPage.tsx`

Endpoint	Method	Purpose	Frontend Component / Page
`/network/telemetry`	`GET`	Deep Packet Inspection (DPI) flow rates	`NetworkIntelligencePage.tsx`
`/identity/users`	`GET`	User identity risk scores and locations	`IdentitySecurityPage.tsx`
`/identity/revoke-session`	`POST`	Terminate session for compromised user	`IdentitySecurityPage.tsx`
`/settings`	`GET/PATCH`	Policy thresholds, auto-mitigation toggles	`SettingsPage.tsx`
`/health/detailed`	`GET`	Subsystem latency and readiness probes	`SystemHealthPage.tsx`

6. Real-Time WebSocket Telemetry & Kernel Verification

WebSocket Subscription

The frontend subscribes to real-time events via `ws://127.0.0.1:8000/api/v1/ws`:

```
{
  "event": "SIMULATOR_SCENARIO_RUN",
  "scenario": "sim-port-scan",
  "threat_id": "THR-SIM-12",
  "rule_id": "FW-RULE-9041",
  "status": "APPLIED_VERIFIED",
  "timestamp": "2026-09-19T14:40:00Z"
}
```

Netlink Kernel Verification Receipts

Every mitigation action (such as dropping an IP or revoking a token) produces an immutable verification receipt:

```
{
  "action": "NFTABLES_DROP_IP",
  "target": "185.220.101.42",
  "status": "APPLIED_VERIFIED",
  "verification_socket": "Linux Kernel Netlink AF_NETLINK",
  "packets_dropped": 14280,
  "bypass_count": 0
}
```

7. Cyber Attack Knowledge Graph & GNN Topology

CyberGuard models attack campaigns as heterogeneous graphs to uncover multi-hop attacker infrastructure:

```
[Threat Actor: Storm-0501]
       |
       ├──(ACQUIRED_SPOOF)──> [DLT Header: OKCL-GOV] ──(TRANSMITS)──> [SMS: Scholarship KYC S
       |                                                                        |
       |                                                                (CONTAINS_LINK)
       |                                                                        ▼
[MITRE T1566: Phishing]                                          [Domain: okcl-scholarship.porta
                                                                        |
                                                                (RESOLVES_DNS_A)
                                                                        ▼
[Victim: deepak.mohanty] ──(IMPOSSIBLE_TRAVEL_SESSION)─────────> [IP: 103.212.45.19 (Frankfurt
                                                                        ▲
                                                                        |
                                                              (ENFORCES_NFTABLES_DROP)
                                                                        |
                                                             [Adaptive FW: FW-RULE-DROP-9
```

Accessible via the interactive canvas in `Intelligence > Attack Graph (GNN)` (`AttackGraphExplorer.tsx`).

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8. Quickstart: Local Installation & Execution

Prerequisites

- Python 3.11+
- Node.js 20+ and `npm`
- Windows, Linux, or macOS

1. Backend Setup

```
# Navigate to backend
cd a:/Cybernates/backend

# Install dependencies
pip install fastapi uvicorn pydantic python-jose argon2-cffi requests

# Launch backend daemon
python -m uvicorn main:app --host 127.0.0.1 --port 8000
```

Backend API Docs will be available at: `http://127.0.0.1:8000/docs`.

2. Frontend Setup

```
# Navigate to frontend
cd a:/Cybernates/frontend

# Install dependencies
npm install

# Start Vite dev server
npm run dev -- --host 127.0.0.1 --port 5173
```

Frontend UI will be live at: `http://127.0.0.1:5173/`.

3. Default Admin Credentials

- **Email:** `elena.vance@cyberguard.internal`
- **Role:** Administrator (Full SOC Clearance)
- **Tenant:** OKCL Core Infrastructure
- *Quick-Login:* Click the **Elena Vance** card on the login window to autofill.

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9. Hackathon Judge Demonstration Runbook

Follow this exact 5-minute sequence when presenting to the judges:

1. **Overview Dashboard (`/overview`):**
2. Show the **92% Nominal SOC Posture**.
3. Point out the **Multi-Source Threat & Impersonation Engine** launcher banner.
4. **Multimodal & Deepfake Threat Lab (`/multimodal`):**
5. **SMS / Smishing:** Run analysis on fake OKCL scholarship grant with spoofed DLT header `OKCL-STUDENT`. Show the 96% risk score and explainable AI breakdown.
6. **QR Quishing:** Show decoded URL payload `okcl-scholarship.portal-claim.online` and OCR notice extraction.
7. **Voice Deepfake:** Play the Managing Director voice memo. Point to the **8,150 Hz vocoder cutoff** and robotic flat pitch line.
8. **Video Deepfake:** Scrub the 120-frame timeline. Point to the purple **facial seam heatmap** around the jawline.
9. **Impersonation Matrix:** Show side-by-side verification: Verified Official (`okcl.org`) vs. Detected Imposter.
10. **Attack Graph & GNN (`/graph-intelligence`):**
11. Show multi-hop correlation: Actor $\rightarrow$ Spoofed Header $\rightarrow$ SMS $\rightarrow$ Phishing Domain $\rightarrow$ IP $\rightarrow$ Victim Session $\rightarrow$

Firewall Rule.

12. Switch to **Multi-Model Training Specs** tab to show KNN, CNN, GNN, and Qwen LoRA mathematical formulas.
13. **Attack Simulator Sandbox (`/simulator`):**
14. Run **Scenario 1: Distributed TCP Port Sweep** or **Scenario 5: Audio Deepfake CEO Fraud**.
15. Watch live terminal execution and automatic `nftables` rule synthesis.
16. **Adaptive Firewall (`/firewall`):**
17. Inspect the newly synthesized drop rule with its 60-minute TTL countdown timer and Netlink kernel verification receipt.